

Week 5 In-Class Worksheet

Hypothesis Testing

ANSWER KEY

Course: H524 Introduction to Biostatistics

Topic: Hypothesis Testing for Population Means

1 Problem 1: Two-Sided t-test

A researcher claims that the average body temperature is not 98.6°F. A sample of 25 healthy adults has a mean temperature of 98.4°F with a standard deviation of 0.8°F. Test the claim at the 0.05 significance level.

Given: $n = 25$, $\bar{x} = 98.4^\circ\text{F}$, $s = 0.8^\circ\text{F}$, $\alpha = 0.05$

Part A: State the null and alternative hypotheses.

Answer:

$$H_0 : \mu = 98.6 \text{ (population mean temperature is } 98.6^\circ\text{F)}$$

$$H_A : \mu \neq 98.6 \text{ (population mean temperature is NOT } 98.6^\circ\text{F)}$$

This is a **two-sided test** because we're testing whether the mean differs in either direction.

Part B: Calculate the standard error and t-statistic.

Answer:

Standard error:

$$SE = \frac{s}{\sqrt{n}} = \frac{0.8}{\sqrt{25}} = \frac{0.8}{5} = 0.16$$

t-statistic:

$$t = \frac{\bar{x} - \mu_0}{SE} = \frac{98.4 - 98.6}{0.16} = \frac{-0.2}{0.16} = -1.25$$

Part C: Find the p-value.

Answer:

Degrees of freedom: $df = n - 1 = 25 - 1 = 24$

For a two-sided test with $t = -1.25$ and $df = 24$:

p-value = $2 \times P(T < -1.25) = 2 \times 0.1117 = 0.2234$

(Using R: `2 * pt(-1.25, df=24)`)

Part D: Make a decision and state your conclusion in context.

Answer:

Decision: Fail to reject H_0 (since p-value = $0.2234 \geq 0.05$)

Conclusion: There is not sufficient evidence at the 0.05 significance level to conclude that the population mean body temperature differs from 98.6°F. The observed difference (98.4°F vs 98.6°F) could reasonably be due to sampling variability.

2 Problem 2: One-Sided t-test (Upper Tail)

A nutritionist claims that a new diet increases average daily caloric expenditure above the baseline of 260 calories. A study of 40 participants following the diet shows a mean expenditure of 285 calories with a standard deviation of 48 calories. Test this claim at the 0.05 level.

Given: $n = 40$, $\bar{x} = 285$, $s = 48$, $\mu_0 = 260$, $\alpha = 0.05$

Part A: State the hypotheses and identify the type of test.

Answer:

$$H_0 : \mu = 260 \text{ (diet does NOT increase expenditure)}$$

$$H_A : \mu > 260 \text{ (diet DOES increase expenditure)}$$

This is a **one-sided (upper tail) test** because we're only interested in whether the mean *increased*.

Part B: Calculate the standard error and t-statistic.

Answer:

Standard error:

$$SE = \frac{s}{\sqrt{n}} = \frac{48}{\sqrt{40}} = \frac{48}{6.325} = 7.5895$$

t-statistic:

$$t = \frac{\bar{x} - \mu_0}{SE} = \frac{285 - 260}{7.5895} = \frac{25}{7.5895} = 3.294$$

Part C: Find the p-value and make a decision.

Answer:

Degrees of freedom: $df = 40 - 1 = 39$

For a one-sided (upper tail) test with $t = 3.294$ and $df = 39$:

p-value = $P(T > 3.294) = 1 - P(T \leq 3.294) = 0.0011$

(Using R: `1 - pt(3.294, df=39)`)

Decision: Reject H_0 (since p-value = 0.0011 < 0.05)

Part D: Interpret the results in context. What does this tell you about the diet's effect?

Answer:

There is strong evidence at the 0.05 significance level that the new diet increases mean daily caloric expenditure above 260 calories. The p-value of 0.0011 means that if the diet had no effect, we would observe a sample mean this large (or larger) only about 0.11% of the time due to random chance. This provides compelling evidence that the diet is effective.

3 Problem 3: Critical Value Approach

Using the data from Problem 2, verify your conclusion using the critical value approach instead of the p-value approach.

Data: $n = 40$, $\bar{x} = 285$, $s = 48$, $t = 3.294$, $\alpha = 0.05$

Part A: Find the critical value for this one-sided test.

Answer:

For a one-sided (upper tail) test with $\alpha = 0.05$ and $df = 39$:

$$t_{\text{critical}} = 1.6849$$

(Using R: `qt(0.95, df=39)`)

Note: We use 0.95 (not 0.975) because this is a one-sided test.

Part B: Compare the t-statistic to the critical value and make a decision.

Answer:

Comparison:

- $t_{\text{statistic}} = 3.294$
- $t_{\text{critical}} = 1.6849$
- $t_{\text{statistic}} > t_{\text{critical}}$ ($3.294 > 1.6849$)

Decision: Reject H_0

The test statistic falls in the rejection region (beyond the critical value), so we reject the null hypothesis.

Part C: Compare this conclusion to the p-value approach from Problem 2. Do they agree?

Answer:

Yes, both methods give the same conclusion!

- **p-value approach:** Reject H_0 ($p = 0.0011 < 0.05$)
- **Critical value approach:** Reject H_0 ($t = 3.294 > 1.6849$)

These two approaches are mathematically equivalent and will always lead to the same decision. The p-value approach provides more information (exact strength of evidence), while the critical value approach provides a clear cutoff point for decision-making.

4 Problem 4: Type I and Type II Errors

A pharmaceutical company is testing a new treatment that they hope is more effective than the standard treatment.

Part A: Define a Type I error in the context of this study. What would be the consequence?

Answer:

Definition: A Type I error occurs when we reject H_0 when H_0 is actually true.

In context: We would conclude the new treatment is more effective than the standard when it actually is *not* more effective.

Consequence:

- Approve and adopt an ineffective new treatment
- Waste resources on inferior or equivalent treatment
- Patients don't benefit from the change
- Opportunity cost of not pursuing better alternatives
- Potential harm if new treatment has worse side effects

Part B: Define a Type II error in the context of this study. What would be the consequence?

Answer:

Definition: A Type II error occurs when we fail to reject H_0 when H_A is actually true.

In context: We would conclude the new treatment is *not* more effective than the standard when it actually *is* more effective.

Consequence:

- Fail to adopt a beneficial new treatment
- Patients continue with inferior treatment
- Miss opportunity to improve health outcomes
- Lose competitive advantage if competitors discover the treatment
- May abandon promising research direction

Part C: Which type of error do you think is more serious in medical research? Explain your reasoning.

Answer:

Generally, Type II errors are considered more serious in medical research.

Reasoning:

- **Type II error:** Patients are directly harmed by not receiving a better treatment. This violates the medical principle of beneficence (doing good).
- **Type I error:** While wasteful, further studies and monitoring would likely detect the error. The scientific process is self-correcting through replication.
- In medicine, missing an effective treatment (Type II) often has greater ethical consequences than temporarily adopting an ineffective one (Type I).

However, the answer depends on context:

- If the new treatment is expensive or has serious side effects, Type I might be worse
- If the disease is life-threatening and current treatment is poor, Type II is definitely worse
- Risk tolerance varies by situation

5 Problem 5: Statistical Power and Sample Size

Answer the following conceptual questions about statistical power.

Part A: How does increasing the significance level (e.g., from 0.05 to 0.10) affect the probability of Type I and Type II errors? What happens to statistical power?

Answer:

Effect of increasing α (e.g., 0.05 $\rightarrow$ 0.10):

- **P(Type I Error) INCREASES**
 - By definition: $P(\text{Type I}) = \alpha$
 - Larger α = more willing to reject H_0
 - More false positives
- **P(Type II Error) DECREASES**
 - Less stringent threshold makes it easier to detect real effects
 - Fewer false negatives
- **Statistical Power INCREASES**
 - Power = $1 - P(\text{Type II})$
 - Lower Type II error $\rightarrow$ Higher power
 - Better ability to detect true effects when they exist

Trade-off: Increasing α makes us more likely to detect real effects (good) but also more likely to falsely claim effects that don't exist (bad).

Part B: How does increasing sample size affect Type I error, Type II error, and statistical power? Explain the mechanism.

Answer:

Effect of increasing sample size n :

- **P(Type I Error) UNCHANGED**
 - Always equals α regardless of sample size
 - We control Type I error by choosing α
- **P(Type II Error) DECREASES**
 - Larger samples give more precise estimates
 - Easier to detect real effects
 - Fewer false negatives
- **Statistical Power INCREASES**
 - Power = $1 - P(\text{Type II})$
 - Better ability to detect true effects

Mechanism:

$$SE = \frac{s}{\sqrt{n}} \quad \Rightarrow \quad \text{Larger } n \Rightarrow \text{Smaller SE}$$

$$t = \frac{\bar{x} - \mu_0}{SE} \quad \Rightarrow \quad \text{Smaller SE} \Rightarrow \text{Larger } |t|$$

Larger $|t| \Rightarrow$ Smaller p-values $\Rightarrow$ Easier to reject H_0 when H_A is true

Example: With $s = 10$:

- $n = 20 \rightarrow SE = 2.2361$
- $n = 80 \rightarrow SE = 1.1180$

Quadrupling the sample size cuts the SE in half, making the t-statistic twice as large (for the same difference in means).

6 Problem 6: Comprehensive Integration Problem

A pharmaceutical company is testing whether a new sleep medication reduces the average sleep time below the standard 8 hours. They conduct a study with 50 patients and find a mean sleep time of 7.8 hours with a standard deviation of 1.2 hours. Test the hypothesis at the $\alpha = 0.01$ level.

Given: $n = 50$, $\bar{x} = 7.8$ hours, $s = 1.2$ hours, $\alpha = 0.01$

Part A: State the null and alternative hypotheses. Is this a one-sided or two-sided test?

Answer:

$$H_0 : \mu = 8 \text{ hours (drug does NOT reduce sleep time)}$$

$$H_A : \mu < 8 \text{ hours (drug DOES reduce sleep time)}$$

This is a **one-sided (lower tail) test** because we're only interested in whether sleep time *decreased*.

Part B: Calculate the standard error.

Answer:

$$SE = \frac{s}{\sqrt{n}} = \frac{1.2}{\sqrt{50}} = \frac{1.2}{7.071} = 0.1697$$

Part C: Calculate the t-statistic.

Answer:

$$t = \frac{\bar{x} - \mu_0}{SE} = \frac{7.8 - 8.0}{0.1697} = \frac{-0.2}{0.1697} = -1.1785$$

Part D: Find the p-value.

Answer:

Degrees of freedom: $df = 50 - 1 = 49$

For a one-sided (lower tail) test with $t = -1.1785$ and $df = 49$:

p-value = $P(T < -1.1785) = 0.1221$

(Using R: `pt(-1.1785, df=49)`)

Part E: Make a decision at the $\alpha = 0.01$ level.

Answer:

Decision: Fail to reject H_0

Reasoning: p-value = $0.1221 \geq 0.01$ (not significant at the 0.01 level)

Part F: Write a complete interpretation of your results in the context of the sleep study.

Answer:

There is not sufficient evidence at the 0.01 significance level to conclude that the new sleep medication reduces mean sleep time below 8 hours. While the sample mean (7.8 hours) is below 8 hours, this difference is not statistically significant at the stringent 0.01 level. The observed difference could reasonably be attributed to random sampling variability rather than a true effect of the medication.

Note: The p-value of 0.1221 means that if the drug had no effect (true mean = 8 hours), we would observe a sample mean this low (or lower) about 12% of the time due to chance alone. This is not strong enough evidence to conclude the drug works.

Part G: Verify your conclusion using the critical value approach.

Answer:

For a one-sided (lower tail) test with $\alpha = 0.01$ and $df = 49$:

$$t_{\text{critical}} = -2.4049$$

(Using R: `qt(0.01, df=49)`)

Comparison:

- $t_{\text{statistic}} = -1.1785$
- $t_{\text{critical}} = -2.4049$
- $t_{\text{statistic}} > t_{\text{critical}}$ (i.e., $-1.1785 > -2.4049$)

The test statistic is NOT in the rejection region (it's not beyond the critical value), so we **fail to reject** H_0 .

This confirms our conclusion from the p-value approach. Both methods agree: there is insufficient evidence that the medication reduces sleep time.

Reflection Questions - Sample Answers

1. Explain the difference between statistical significance and practical significance. Can a result be statistically significant but not practically important?

Sample Answer:

Statistical significance means the observed effect is unlikely to be due to chance alone ($p\text{-value} < \alpha$). It tells us the result is *real* (not random), but not how *large* or *important* it is.

Practical significance (or clinical significance) means the effect is large enough to matter in the real world. It addresses whether the difference is meaningful for decision-making.

Yes, a result can be statistically significant but not practically important!

Example: With a sample of 10,000 people, we find that a new diet reduces weight by 0.1 pounds on average ($p < 0.001$). This is highly statistically significant, but the actual weight loss is so small it's practically meaningless. No one would adopt a restrictive diet to lose 1/10th of a pound.

Key insight: Large samples can detect tiny, trivial differences as “significant.” Always consider effect size, not just p-values.

2. Why do we typically use $\alpha = 0.05$ as the significance level? What would happen if we used $\alpha = 0.10$ or $\alpha = 0.01$ instead?

Sample Answer:

Why 0.05?

- Historical convention (R.A. Fisher's suggestion from the 1920s)
- Balances Type I and Type II errors reasonably well
- Widely accepted in most fields, making results comparable
- No deep mathematical reason - it's a social convention

Using $\alpha = 0.10$ (more lenient):

- **Pro:** Higher power, easier to detect real effects
- **Con:** More false positives (10% vs 5%)
- **Use when:** Exploratory research, preliminary screening, cost of Type II error is high

Using $\alpha = 0.01$ (more stringent):

- **Pro:** Fewer false positives (1% vs 5%), stronger evidence
 - **Con:** Lower power, harder to detect real effects
 - **Use when:** High stakes decisions, confirmatory research, cost of Type I error is high
- Bottom line:** Choice of α should match the consequences of errors in your specific context.

3. When would you use a one-sided test versus a two-sided test? Give an example of each.

Sample Answer:

Two-sided test (most common):

- Use when you want to detect differences in *either direction*
- Testing whether something is *different*, not specifically higher or lower
- More conservative and generally preferred
- $H_A : \mu \neq \mu_0$

Example: Testing whether a new manufacturing process changes the average product weight. We care if it's heavier *or* lighter than the target.

One-sided test (less common):

- Use when you're only interested in one direction
- You have strong prior reason to expect a specific direction
- Differences in the opposite direction would be treated the same as no difference
- $H_A : \mu > \mu_0$ or $H_A : \mu < \mu_0$

Example: Testing whether a new safety feature *reduces* injury rates. We only care about reduction - an increase would still mean we don't adopt the feature.

Important: Choose one-sided vs two-sided *before* seeing the data. Switching after seeing results inflates Type I error.

4. What is the relationship between confidence intervals and hypothesis testing?

Sample Answer:

Confidence intervals and hypothesis tests are closely related:

For a two-sided test at level α :

- If the null value μ_0 is *outside* the $(1 - \alpha)100\%$ CI, reject H_0
- If the null value μ_0 is *inside* the $(1 - \alpha)100\%$ CI, fail to reject H_0

Example: Testing $H_0 : \mu = 100$ at $\alpha = 0.05$

- Calculate 95% CI: (105, 115)
- Since 100 is outside this interval, reject H_0 at the 0.05 level
- The CI gives the same conclusion as a formal hypothesis test

Advantages of CIs over hypothesis tests:

- Shows the *magnitude* of the effect (effect size), not just significance
- Gives a range of plausible values
- Shows precision of the estimate
- More informative than just “reject” or “fail to reject”

Best practice: Always report confidence intervals along with hypothesis test results!